

بدون

Idea: Randomly Construct A . & prove that $RIP(A, 2s, \frac{1}{3})$ is satisfied with high prob. as long as $n = O(s \log d/s)$.

Assumption:

$A_{ij} \stackrel{iid}{\sim} \text{Rad.}$

$$|S| \leq s \quad \|A_s^T A_s / n - I\|_2 = \sup_{\|v\|_2=1} |v^T (\overbrace{A_s^T A_s}^{M_s} / n - I) v|$$

$$= \sup_{\|v\|_2=1} \left| \frac{(A_s v)^T (A_s v)}{n} - \underbrace{\|v\|_2^2}_1 \right|$$

$$\frac{1}{n} \left\| \begin{bmatrix} \overbrace{-a_1-}^{A_s} \\ \vdots \\ -a_n- \end{bmatrix} v \right\|_2^2 = \frac{1}{n} \left\| \begin{bmatrix} \langle a_1, v \rangle \\ \vdots \\ \langle a_n, v \rangle \end{bmatrix} \right\|_2^2 = \frac{1}{n} \sum_{i=1}^n \langle a_i, v \rangle^2$$

$$= \underbrace{\left| \frac{1}{n} \sum_{i=1}^n \langle a_i, v \rangle^2 - 1 \right|}_{Z} \rightarrow \mathbb{E}\{Z\}$$

$$\mathbb{E} \left\{ \frac{(A_s v)^T (A_s v)}{n} \right\} = \mathbb{E} \left\{ v^T \frac{A_s^T A_s}{n} v \right\} = v^T \underbrace{\mathbb{E} \left\{ \frac{A_s^T A_s}{n} \right\}}_I v = 1$$

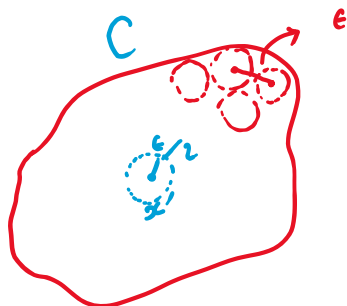
$$\langle a_i, v \rangle \sim \text{subG}(\sqrt{v_1^2 + \dots + v_s^2}) = \text{subG}(1)$$

$$\langle a_i, v \rangle^2 \sim \text{subE}(c')$$

$$\forall v \in S^{s-1} \quad \mathbb{P} \left(\left| \frac{1}{n} \sum_{i=1}^n \langle a_i, v \rangle^2 - 1 \right| \geq t \right) \leq 2e^{-cnt^2} \quad t \leq t_0$$

$$(*) \quad \mathbb{P} \left(\sup_{\|v\|=1} |Z_v - \mathbb{E}(Z_v)| \geq t \right) = \mathbb{P} \left(\exists v \|v\|=1: |Z_v - \mathbb{E}(Z_v)| \geq t \right)$$

Packing

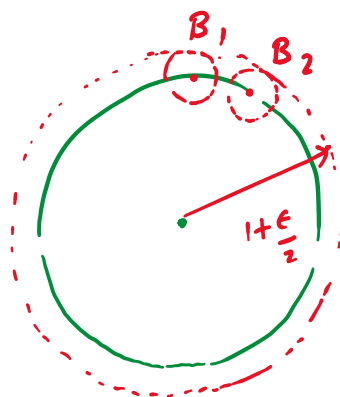


radius $\epsilon/2$ at least
centers are ϵ away

Maximal number of balls

If I choose any arbitrary point x in this set C , $\exists$ one ball whose center has $\text{dist} < \epsilon$ to the center.

Now let's calculate the size of packing number N for the unit sphere.



$$\bigcup_{i=1}^N B_i(\epsilon/2) \subseteq B(1 + \epsilon/2)$$

$$\text{vol} \leq \text{vol}(B(1 + \epsilon/2))$$

$$\sum \text{vol}(B_i) \leq (1 + \epsilon/2)^d \text{vol}(B(1))$$

$$N (\epsilon/2)^d \text{vol}(B(1))$$

if

$$N \leq \left(\frac{2}{\epsilon} + 1\right)^d = \left(\frac{2 + \epsilon}{\epsilon}\right)^d \leq \left(\frac{3}{\epsilon}\right)^d$$

$$\mathbb{P}\left(\sup_{\|v\|_2=1} |v^T M_S v| \geq t\right) \leq \mathbb{P}\left(\max_{v \in \text{Packing } S^{s-1}} |v^T M_S v| \geq t'\right)$$